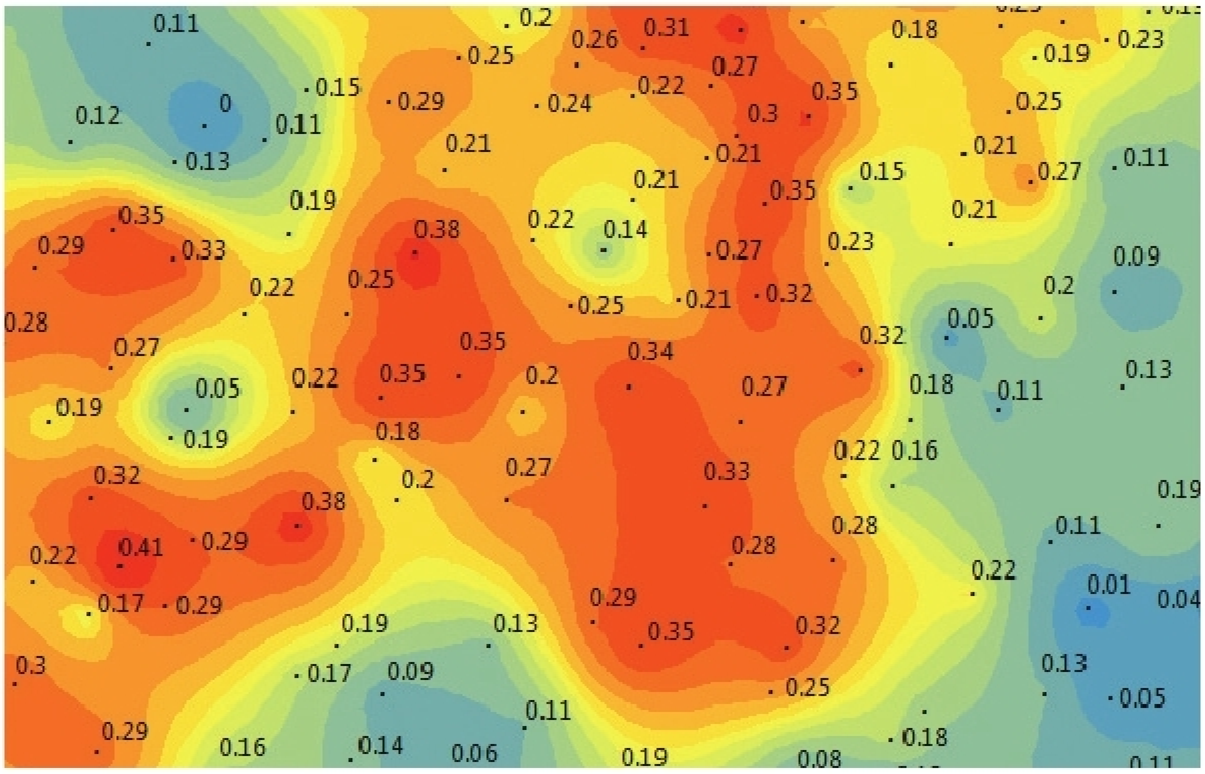


Figure 2: Empirical Bayesian Kriging

Figure 3: K-means clustering

K-means algorithm first segments the image for further analysis. Next, each cluster is assigned a land cover class.

However, GIS can use clustering in other unique ways. For example, it can be used for crime mapping—the data points could represent crime locations that can be clustered as spots of crime.

## AI applications in GIS

The applications of artificial intelligence in GIS are in a number of sectors, including those that use location and GIS. Ride-sharing companies, logistics, agriculture, surveying, and infrastructure are some of the prominent examples.

**Predictions and remote monitoring:** AI based GIS applications can handle complex weather and climate imagery data patterns that humans can't process on a large scale and in real-time. Such GIS applications can result in solutions for problems arising from climate change, air pollution, water pollution and forest management. This framework can optimise the use of land data, agriculture data, regional based crop data, etc, to maximise the economic benefits for society.

**Internet of Things (IoT):** Almost every connected device that uses GIS application software can use artificial intelligence as a platform to predict, adapt, learn and make decisions for end users. The neural geospatial data system can be the core engine for self-driving vehicles and drones to adapt and manipulate an environment in real-time, using all kinds of GIS data.

**Geographic consumer behaviour:** In today's competitive world, understanding the consumer is the focus of any organisation. AI based structures can learn and predict consumer behaviour using geographic data and come up with specialised regional solutions. Ride-sharing companies like Uber, Ola, etc, can take feedback from customers and process the data to find out the density of cars and the availability of drivers. In logistics and supply chains, artificial intelligence can plug the gaps and gather more accurate location information that can streamline product delivery and save time.

Overall, in the realm of business, artificial intelligence based systems can create an ecosystem around various GIS applications that would substantially improve planning, resource allocation and decision-making – predicting the surge in demand and supply, identifying the prospects of high and low margins, multiplying supply chain efficiency, and optimising service delivery. The scope is endless. END

**By: Miren Karamta**

The author is a project scientist and IT systems manager at the Bhaskaracharya Institute for Space Applications and Geo-informatics (BISAG), Gandhinagar, Gujarat. You can reach him via email at *mirenkaramta@yahoo.com.* LinkedIn: *https://in.linkedin.com/in/miren-karamta-2b929122*